THE QUOT FUNCTOR IS AN ALGEBRAIC SPACE, LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-quot-sheaf (canonical tag 09TT). cited at proof line 2540. The Quot functor is an fppf sheaf.
- Section algebraic-section-split (canonical tag 04SU). cited at proof line 2544. Passing from the sheaf to the stack in groupoids.
- Proposition algebraic-proposition-algebraic-stack-no-automorphisms (canonical tag 04SZ). cited at proof lines 2545-2546. Reduction to algebraicity of the Quot stack.
- Theorem theorem-coherent-algebraic-general (canonical tag 09DS). cited at proof line 2557. Cohstack_{X/B} is an algebraic stack.
- Lemma algebraic-lemma-representable-morphism-to-algebraic (canonical tag 05UM). cited at proof lines 2559-2560. Suffices to show the 1-morphism is representable by algebraic spaces.
- Lemma stacks-lemma-2-fibre-product-gives-stack-in-setoids (canonical tag 05UI). cited at proof lines 2576 and 2579. Z is a stack in setoids; its sheaf of sets.
- Lemma stacks-lemma-when-stack-in-sets (canonical tag 0430). cited at proof line 2580. Identifying the sheaf of sets attached to Z.
- Lemma spaces-flat-lemma-F-surj-open (canonical tag 09TP). cited at proof line 2584. Surjectivity locus is open: Z is an open subspace.
- Proposition proposition-hom (canonical tag 08K6). cited at proof line 2587. The ambient algebraic space Hom(F_T, G).

Source reference. Original lemma 09TT: sources/stacks-chow/tags.txt, source lines 12401–12401. Cited at proof line 2540: sheaf property of the Quot functor; load-bearing.

Source reference. Original section 04SU: sources/stacks-chow/tags.txt, source lines 6232–6232. Cited at proof line 2544: passing from the sheaf to the associated stack in groupoids; load-bearing.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original proposition 04SZ: sources/stacks-chow/tags.txt, source lines 6237–6237. Cited at proof lines 2545-2546: reduces to showing the Quot stack is algebraic; load-bearing.

Source reference. Original theorem 09DS: sources/stacks-chow/tags.txt, source lines 11875–11875. Cited at proof line 2557: Cohstack_{X/B} is an algebraic stack; load-bearing.

Source reference. Original lemma 05UM: sources/stacks-chow/tags.txt, source lines 7521–7521. Cited at proof lines 2559-2560: suffices to show the 1-morphism is representable by algebraic spaces; load-bearing.

Source reference. Original lemma 05UI: sources/stacks-chow/tags.txt, source lines 7517–7517. Cited at proof lines 2576 and 2579: the fibre product Z is a stack in setoids and corresponds to a sheaf; load-bearing twice.

Source reference. Original lemma 0430: sources/stacks-chow/tags.txt, source lines 5347–5347. Cited at proof line 2580: identifying the sheaf of sets attached to Z; load-bearing.

Source reference. Original lemma 09TP: sources/stacks-chow/tags.txt, source lines 12397–12397. Cited at proof line 2584: Z is representable by an open subspace of the Hom space; load-bearing.

Source reference. Original proposition 08K6: sources/stacks-chow/tags.txt, source lines 10874–10874. Cited at proof line 2587: the ambient algebraic space Hom(F_T, G); load-bearing.

Proposition 0.2. *Let S be a scheme. Let $f : X \to B$ be a morphism of algebraic spaces over S. Let $\mathcal{F}$ be a quasi-coherent sheaf on X. If f is of finite presentation and separated, then $\mathrm{Quot}_{\mathcal{F}/X/B}$ is an algebraic space. If $\mathcal{F}$ is of finite presentation, then $\mathrm{Quot}_{\mathcal{F}/X/B} \to B$ is locally of finite presentation.*

Proof. By Lemma 09TT we have that $\mathrm{Quot}_{\mathcal{F}/X/B}$ is a sheaf in the fppf topology. Let $Quot_{\mathcal{F}/X/B}$ be the stack in groupoids corresponding to $\mathrm{Quot}_{\mathcal{F}/X/S}$, see Algebraic Stacks, Section 04SU. By Algebraic Stacks, Proposition 04SZ it suffices to show that $Quot_{\mathcal{F}/X/B}$ is an algebraic stack. Consider the 1-morphism of stacks in groupoids

$$Quot_{\mathcal{F}/X/S} \longrightarrow \mathcal{C}oh_{X/B}$$

on $(Sch/S)_{fppf}$ which associates to the quotient $\mathcal{F}_T \to \mathcal{Q}$ the module $\mathcal{Q}$. By Theorem 09DS we know that $\mathcal{C}oh_{X/B}$ is an algebraic stack. By Algebraic Stacks, Lemma 05UM it suffices to show that this 1-morphism is representable by algebraic spaces.

Let T be a scheme over S and let the object $(h, \mathcal{G})$ of $\mathcal{C}oh_{X/B}$ over T correspond to a 1-morphism $\xi : (Sch/T)_{fppf} \to \mathcal{C}oh_{X/B}$. The 2-fibre product

$$\mathcal{Z} = (Sch/T)_{fppf} \times_{\xi, \mathcal{C}oh_{X/B}} Quot_{\mathcal{F}/X/S}$$

is a stack in setoids, see Stacks, Lemma 05UI. The corresponding sheaf of sets (i.e., functor, see Stacks, Lemmas 05UI and 0430) assigns to a scheme T'/T the set of surjections $u : \mathcal{F}_{T'} \to \mathcal{G}_{T'}$ of quasi-coherent modules on $X_{T'}$. Thus we see that $\mathcal{Z}$ is representable by an open subspace (by Flatness on Spaces, Lemma 09TP) of the algebraic space $Hom(\mathcal{F}_T, \mathcal{G})$ from Proposition 08K6. □